Chuang Yang

PhD Student of Control Science and Engineering,
Nanjing University of Aeronautics and Astronautics, China.

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✧ Contact

E-mail: yangchuang@nuaa.edu.cn, yang_ch126@126.com
Tel.: +86 166-5161-9089
Skype: live:88d473bdc4f5f860
LinkedIn profile: linkedin.com/in/chuang-yang-02704b197
Address: College of Automation Engineering,
Nanjing University of Aeronautics and Astronautics,
29 Jiangjun Road, Nanjing 211106, China.

✧ Personal Details

Given/Family Name: Chuang Yang | **Gender**: Male | **Birth**: 1992 | **Marital status**: Single
Current status: 3rd-year PhD Student at NUAA (5-year full-time doctor's degree program).
Language: Mandarin Chinese, English (professional proficiency).
Personal Homepage: www.yangchuang.link

✧ Current Research

- **Major**: Control Science and Engineering
- **Research interests:**
  Swarm perception, awareness and navigation in Robotics and Neuroscience; Neural basis of spatial cognition and navigation; Self-organizing behavior and complex dynamical processes in biological systems; Cooperative Navigation; Integrated Navigation System (INS/GNSS/Magnetic); Neuromorphic Swarm; Bio-inspired self-organizing and collective robots.
- **PhD topic:** Brain-inspired swarm navigation for autonomous collective robots

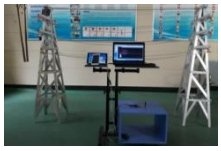
Abstract: This project (www.neuromorphicswarm.com) will fuse social perception/communication/ collective behavior of animal groups, neural basis of spatial representations of self and other in the hippocampus, and the decentralized cooperative navigation in robotics by employing neuromorphic computing modle (e.g. CANN, ML, SNN) to develop a new class of brain-inspired swarm perception, awareness and navigation architecture——towards neuromorphic devices——in large-scale 3D environments for Neuromorphic-Swarm-Robot that navigate in 3D space as social animals do.

- **Journal Reviewer :**
  Chinese Journal of Aeronautics.
- **Professional association:**
  Member of Royal Institute of Navigation, Member of Chinese Society of Aeronautics and Astronautics, Member of Chinese Association for Artificial Intelligence, Member of Chinese Association of Automation.

✧ Skill Profile

- **Programming**
  MATLAB, C, LabVIEW, Python
- **Web Design**
  Hugo, Github
- **Open Source Program**
  RatSLAM: a brain-based SLAM system
  NeuroSLAM: a brain-inspired SLAM system for 3D environments
  SwarmLab: an open source MATLAB Drone Swarm Simulator
  GNSS/INS: an Integrated Navigation System Simulator (designed by my group).
- **Embedded Hardware**
  Arduino, STM32
- **Design of Electronic System**
  (***I became interested in electronic products in childhood, greatly influenced by my parents who were all maintanance workers for electronic devices, such as television, amplifier, etc.***)
  Two seclected designs of electronic system:

IMU/Magnetic Sensors-based Integrated Navigation System of Measurement While Drilling
Third Prize in the 4th Petroleum Equipment Innovative Design Competition for China postgraduate (***team leader***), 09/2017.



IMU-based On-line Monitoring System of Transmission Line Galloping
Second Prize in the 12th Electronic Design Competition for China postgraduate (***team leader***), 07/2017.

✧ Education Experiences

- **04/2018—present**
  Nanjing University of Aeronautics and Astronautics, China
  PhD student of Control Science and Engineering
  Supervisor: Prof. Jianye Liu
  Co-supervisor: Prof. Zhi Xiong
  Thesis Topic: *Brain-inspired swarm navigation for autonomous collective robots*
- **07/2020—07/2020**

Certificate of Completion of Neuromatch Academy (an online school for Computational Neuroscience)
Interactive student of Computational Neuroscience
Mentor: Dr. Torres Gelsy (Universtity of Pittsburgh)
Course Project: *How does task related activity interact with intrinsic activity of the brain*?

- **09/2015—12/2017**
  Henan Polytechnic University, China
  Master of Control Science and Engineering (***graduating half a year in advance***)
  Supervisor: Dr. Jinxian Yang
  Final thesis: *Research on INS/magnetic integrated navigation technique for measurement while drilling*
- **09/2011—07/2015**
  Henan Institute of Science and Technology, China
  Bachelor of Electrical Engineering and Automation (GPA: 3.68/4.0)

✧ Seclected Project Experiences

- **Brain-inspired navigation & swarm navigation**

1. China, National Natural Science Foundation of China, Grant No. 61873125. **Brain-inspired intelligent cooperative navigation for autonomous collective robots,** 01/2019-12/2022. (main participant, my PhD topic).
2. China, Shanghai Aerospace Science and Technology Innovation Fund, Grant No. SAST2019-085. **Brain-inspired positioning technology for intelligent spacecraft**, 12/2019-12/2021. (main participant).

- **Integrated navigation**

3. China, National Natural Science Foundation of China, Grant No. U1404510. **Key technologies of MEMS inertial navigation for measurement while drilling**, 01/2015-12/2017. (main participant)
4. China, Key Science and Technology Program of Henan Province, Grant No. 172102210289. **Research on on-line monitoring system of transmission line galloping based on inertial technology**, 09/2016-09/2017. (main participant)

✧ Publications

- **Brain-inspired navigation & swarm navigation**

1. **Chuang Yang**, Zhi Xiong, Jianye Liu, Lijun Chao, Yudi Chen. **Brain-inspired metric navigation in large-scale 3D environments based on model of multi-scale grid cells**[J]. IEEE Transactions on Cognitive and Developmental Systems (Under review, 2020.08).
2. **Chuang Yang**, Jianye Liu, Zhi Xiong, Lijun Chao. **Brain-inspired vector navigation method based on model of multi-scale grid cells**[J]. Journal of Chinese Inertial Technology, 2020, 28(02):179-185. (EI, in Chinese)
3. **Chuang Yang**, Jianye Liu, Zhi Xiong. **A new paradigm of brain-inspired navigation integrating perception and decision**[EB/OL]. article.xuexi.cn, 2020.03.04. (in Chinese)
4. **Chuang Yang**, Jianye Liu, Zhi Xiong, Jun Xiong. **Brain-inspired navigation technology integrating perception and action decision: A review and outlook**[J]. Acta Aeronautica et Astronautica Sinica, 2020, 41(1):323280. doi:10.7527/S1000-6893.2019.23280. (EI, in Chinese)
5. Jianye Liu, **Chuang Yang**, Zhi Xiong, Jizhou Lai, Jun Xiong. **Attractor neural network-based brain-inspired navigation technology**[J]. Navigation Positioning & Timing, 2019, 6(5): 52-60. (in Chinese)
6. **Chuang Yang**, Jianye Liu. **Brain-inspired navigation technology and its applications in unmanned autonomous system**[J]. Naas & Inertal Technology, 2019, 200+201:151-152. (in Chinese)

- **Integrated navigation**

7. Jinxian Yang, **Chuang Yang**, Shuanglei Li, et al. **Vibration analysis of transmission lines based on MIMU**[J]. Journal of Vibration and Shock, 2018, 37(11): 230-236. (EI, in Chinese)
8. Jinxian Yang, **Chuang Yang**, Jinbao Jiang. **A robust mag/INS-based orientation estimation algorithm for measurement while drilling**[J]. IEEE Sensors Journal, 2017, 17(23): 7971–7980. (SCI)
9. Jinxian Yang, **Chuang Yang**. **MEMS gyro signal reconstruction via layerwise and adaptive MP**[J]. Optics and Precision Engineering, 2017, 25(12): 3160-3168. (EI, in Chinese)
10. Jinxian Yang, **Chuang Yang**, Zhitao Jiang. **Adaptive extraction of gravity information for drilling tool with magnetic and inertial integrated system**[J]. Journal of china coal society, 2017, 42(12): 3331-3337. (EI, in Chinese)
11. Jinxian Yang, **Chuang Yang**. **Signal extraction of mems gyro utilized in MWD based on sparse representation**[J]. Chinese Journal of Sensors & Actuators, 2017, 30(11): 1677-1683. (in Chinese)
12. Jinxian Yang, **Chuang Yang**. **Gyro drift estimation and compensation based on gravity quaternion**[J]. Chinese Journal of Sensors and Actuators, 2017, 30(8): 1187-1192. (in Chinese)

✧ Patents

1. **Chuang Yang**, Zhi Xiong, Jianye Liu, et al. **A brain-inspired navigation method in 3D environment based on path integration of multi scale grid cells**[P]. China Invention Patent, (2020-08, Under review)

✧ Honors and Awards

10/2017: National Scholarship for Postgraduate

09/2017: Third Prize in the 4th Petroleum Equipment Innovative Design Competition for China postgraduate, ***INS/Magnetic-based integrated navigation system of measurement while drilling***. (***team leader***)

07/2017: Second Prize in the 12th Electronic Design Competition for China postgraduate, ***INS-based on-line monitoring system of transmission line galloping***. (***team leader***)

09/2015-01/2018: Full scholarship for Postgraduate

11/2013: National Scholarship for Encouragement

03/2013: Third-level Certificate for National Computer

11/2012: National Scholarship for Encouragement

09/2011-07/2015: The First Prize Scholarship for four years